



وفاة جرينتك
WafraGreentech

What changed in the mockup

Every change made to the Wafra Farm App UI mockup at version 1.5.6, round by round, in the order the comments arrived — the recorded call of 22 August and the four rounds of comment on the deck that followed it.

The build number is held at 1.5.6 on instruction: one deck, one number, while the review cycle is in flight. The requirement set underneath it is at 1.7.

Romain Bigare

Wafra Farm App · UI mockup v1.5.6 · Build Specification v1.7

What v1.5.4 changed, and where each of it landed

Round 1 of three. The comments on this deck cut six more screens and changed the icons, the plot shapes and the flow diagram; see [Mockup_Changes_v154_round2.md](#). Where the two disagree — the four judgement calls below in particular — the later document wins. The build number stays at 1.5.4 across all three rounds on instruction.

The companion to [Meeting_Comments_v152.md](#), which is the extraction from the call. This is the record of the build: every row in that table, the file it landed in, and the four judgement calls that were mine rather than the review's.

the specification to **1.6** — the first time the spec number has moved since v1.2 was issued as a document. Four earlier rounds redrew screens; this one deleted two whole concepts and changed a third, and no amount of redrawing would have implemented it. Both live in `app/meta.js`.

The five decisions, and what each one cost

1. Tasks are gone; advice is the only unit of work

The deepest change, and the one everything else hangs off. There is no task record, no task list, no assignment and no completion form.

An advice is now in one of four states, and `sentAt` is the whole of the difference between the first two:

STATE	WHAT IT MEANS
open, not sent	the farmer has not decided
open, sent	out with the supervisor, waiting for him to confirm
done	somebody recorded what was actually done, on D7
deferred	ignored or put off; it comes back tomorrow

`groupedTasks()`, `createTask()`, `completeTask()`, `blockTask()`, `assignAllAdvice()`, the `tasks` fixture array, the Tasks tab and WF3.004's badge, `TASK_ICON`, and the `tasks` entitlement key.

`unsentAdvice()`, `can('advice.send')`.

`tools/syntax.sh` now fails the build if the word "task" appears in live code outside a comment, or if any module other than `advice.js` calls `sendAdvice()`. Task management came back once before, one convenience button at a time.

2. The workforce is gone, and the worker role with it

\$5.6 was a directory: worker records, per-person languages, SMS/WhatsApp/push delivery pipes, invitations bound to records, and an identity model joining accounts to people by mobile number. All of it is deleted.

What replaced it is one selector — `supervisorOf(farmId)` — and one rule: work goes to that person as a message with a link they tap when it is done. Nobody below them holds an account.

The **worker role** went with it, which was not asked for in those words but follows from Mark's "there's only going to be one user": with no queue to hold and nothing to mark done, a worker account had nothing in it. The capability matrix ships two roles. A15 redeems an invitation as the farm's supervisor.

fixtures, assignees(), deliveryFor(), identityIds(), attachAccount(), accountForNumber(), the QR sheets, and the worker rows of the capability matrix, the notification categories and the cache/offline tables.

E6 and E7 survive and are **not** tasks — they are field capture, a person standing in a plot recording what the satellite cannot see. They moved to `app/screens/field.js`.

3. Single farm is the product

`router.homeRoute()` decides where Home opens: B1 for an account with more than one farm, B2: <the farm> for everybody else. It is computed rather than stored, so adding a second farm puts the chooser back without anyone remembering to.

Multi-farm stays in the requirements and in the build; it is simply not on the path a single-farm owner walks. Inside a farm, the app-bar title is a picker for accounts that have somewhere to go.

4. B2 and B3 merged

B3 is deleted. B2 is the farm **and** every plot on it: map, one line saying whether anything is urgent, the advice count, then the plots — crops first, tree groups second, under the state filter and the colour key that used to live on B3.

Three blocks came off the same screen in the same round:

- **"Health today"** — plant health, water stress and nutrition read at farm level. Every crop has its own profile, so the farm-level figure averages things that cannot be averaged. The measures are untouched on the plot and the map.
- **Weather** — moved to More, as the new **F15**. Given a screen of its own it prints the whole forecast the plan pays for rather than three columns and a "+11 days".
- **Workforce and Farm diary** — deleted with task management.

5. Trees are tree groups, and they must be surveyed

A tree group is one record per species per farm: `kind: 'trees'`, a species, a tree count, and a `parcels` count saying how many separate pieces of ground its trees stand on. It has no crop cycle — citrus is citrus — and no hand-drawn boundary, because trees are counted individually from the imagery and the count is what the price is worked out from.

Al Kharj North's **twelve date-palm plots are now one group of 7,801 palms**, standing in three places. Al Kharj South has four crop plots and four tree groups, which is why the deck opens B2 on it: farm-1 demonstrates the fold perfectly and the merged screen not at all.

The geometry change is contained: a plot is a list of rings (`plot.patches`) rather than one ring, `plot.geometry` is still the largest of them, and only `ui/map.js` had to learn about it. A group's parcels are drawn with a dashed outline so three shapes read as one holding.

survey and a sentence saying why, not a disabled card to argue with. A12 keeps the answer and the quote; it is the price screen now rather than the question screen.

The reminder that was new

Mark's case: it is January, the tomatoes are off, something else is in, and the satellite cannot name it for about three weeks because it reads a canopy.

A plot carries `harvestDetected0n`. While it is set, the plot has no current crop; the row on B2 says so in red with "Tell us what you planted", and B4 opens on the same question above the imagery rather than below three screens of it. `declareCrop()` clears it and opens a cycle. `plot-23` is in that state in the fixtures.

Everything else, by file

WHAT	WHERE
Urgent-only attention counts	<code>home.js</code> <code>urgentCount()</code> , and the wording of <code>b1.needning</code> / <code>b2.needning</code>
Four tabs, not five	<code>core/capabilities.js</code> <code>tabsFor()</code>
Home opens on the farm for single-farm accounts	<code>core/router.js</code> <code>homeRoute()</code>
Plot rows: name, size, what is growing, urgent flag; no NDVI	<code>home.js</code> <code>plotRow()</code>
Crops and tree groups as two blocks	<code>home.js</code> <code>B2()</code>
Plot list filter — all / urgent / action needed / watch / good	<code>state.ui.plotFilter</code>
Advice filtered by action type, not by assignee	<code>advice.js</code> <code>STATE_FILTERS</code> , <code>state.ui.adviceStateFilter</code>
Send / Ignore before, Record / Take it back after	<code>advice.js</code> <code>adviceCard()</code>
Send everything waiting, and an auto-send switch	<code>advice.js</code> <code>sendAllBar()</code>
Tree group detail: count, parcels, no cycle	<code>plot.js</code> <code>cropHeader()</code>

WHAT	WHERE
Boundary editing and crop cycles refused on a tree group	<code>overlays.js</code> <code>PLOT_MENU</code>
Weather	<code>more.js</code> <code>F15()</code>
Buttons say where they land, for a printed deck	<code>b1.seewhat.dest</code> , <code>b2.advice.dest</code>
Realistic plot geometry	unchanged — the shapes were already jittered parcels; what changed is that a farm now has four of them rather than twelve

Four judgement calls

Each is a small change back if it was read wrongly.

1. **Which four tabs.** The review agreed Tasks goes and that Plots earns a slot, and left the final set to me. Merging B2 and B3 made a Plots tab a second door onto the screen the farmer is already standing on, so the four are **Home, Map, Advice, More** and Home is the plot list. 2. **The worker role is deleted, not just the workforce screens.** Follows from "there's only going to be one user", but it was not said in those words. 3. **B3 is deleted rather than kept as a thinner screen.** "Merge the two" and "make B3 the home screen" are the same instruction from two directions; a surviving B3 would have duplicated B2. 4. **The activity log (F11) stays under More.** The "Farm diary" row on B2 is gone as asked, but F11 is the append-only audit log of WF5.187/WF5.188 rather than the diary the farmer was being offered, and deleting it would have taken `auditlog.view` with it.

Still open

- **One tree group per species, or one for all trees.** Mark is checking with Neil. The build goes per-species: lemon trees get different advice from date palms.
- **Who pays for the farmer's licence,** and pre-registration by phone number against ADAFSA's 25,500-farm list. The existing large-government registration path already covers it and needs no change yet.

Checks

`npm run check` · `npm run smoke` · `npm run deck`

The smoke test walks 579 renders across both roles and now also asserts: every farm has a supervisor to send work to; only the owner may send; an advice card carries Send/Ignore or Record/Take-it-back and never "Mark as complete"; and A9 offers one route for a farm of trees and two for a farm of crops.

The deck is `docs/Wafra_Farm_App_Screens_v1.5.4.pptx` — 62 slides, generated from the running app, so the screenshots cannot drift from the build.

v1.5.4, second and third rounds of comment

The build number stays at **1.5.4** on instruction: one deck, one number, while the review cycle is still in flight. The specification has moved twice underneath it and is at **1.7** — the two answer different questions, which is the point of holding both.

- Round 1, the recorded call: `Mockup_Changes_v154.md`
 - Round 2 and 3, the comments on the deck: this document
-

Round 2 — everywhere

No screen names another screen

The build had begun annotating buttons with where they land — *"Opens your advice list (D1)"*. That belongs on the deck, drawn as arrows, not in an app a farmer uses. Gone.

Lucide, vendored

The ~90 hand-written path strings looked hand-drawn because they were. They come from **Lucide** now, read out of `lucide-static` by `tools/build-icons.mjs` into `app/ui/icons.data.js`. **Not a CDN**, though one was offered: this file is opened from `file://` and photographed by two headless browsers, and a network round trip is a way for any of those to come back with empty boxes. `npm run icons` regenerates.

Every plot is a rectangle

The wobbling five- and six-sided parcels and the centre-pivot circles were making a satellite mockup look like a hand drawing. The farm grid is square too, so parcels are no longer all portrait. The drawing canvas opens on a rectangle.

Six screens deleted

the `FARM_SWITCH` sheet now, carrying Add a farm), **B7** and **B8** (both were the map rebuilt at plot scope), **B9** (replaced by **B13**, the tree group),

Round 3 — screen by screen

SCREEN	WHAT CHANGED
B2	The plot list loses its card: hairlines between rows and nothing else, at <code>--ink-200</code> so they are actually visible without the box round them. The size moves up beside the plot name. Only the chevron navigates — the row itself is not a target, so the crop control beside it can be pressed without opening the plot. "Harvested — tell us what you planted" is "Set the new crop" , one line.
B12 / A9	Nothing is drawn disabled. The route cards greyed out until the farm had a name, which reads as broken rather than as sequenced and leaves the farmer guessing which field unlocks it. The name is still required — pressing a card without one puts the cursor in the name field and says why — but the choice looks like a choice from the moment the screen opens.

SCREEN	WHAT CHANGED
B4	The map pulls back: the pad round a plot scales with the plot rather than sitting at a fixed 20 m, so the boundary is no longer hard against the frame. "What is growing here now" goes from five stacked paragraphs to two lines — the question, then the fact and the button — with the satellite-phenology explanation behind an ①.
A3	Mobile / Email pills , per the example: a filled segmented control at the top, with the form beneath it swapping. It was one form with a "Log in with email and password" link at the bottom, which made one route the screen and the other a footnote a farmer had to read past his own form to find.
A5	Unchanged — the pills were tried here and taken out again. Registering is not a choice between two ways to prove who you are: an account needs both the mobile (it is the account) and the email (it finds a licence bought elsewhere), so a switch over two things you are going to fill in anyway only adds a step.
A12	Retitled "Your survey" . "What we will look for" is one line, and so is its answer. The explanation drops from five paragraphs to four short rows, the update-frequency line is gone, and the warning box is gone — the combined-service note is a sentence, because it is good news about the price rather than a caution.
A10 / A10D	The instruction — thirty-eight words wrapped over three lines above a map the farmer is trying to look at — moves behind a "How to draw this" chip. It is not lost, it is one tap away, and a farmer who has drawn one boundary never has to read it again.

The flow diagram is a tree

Undone in round 4. The tree was built, printed, and taken back out at the next review in favour of a linear strip that follows the section the page is filed under. Kept here because it happened; see [Mockup_Changes_v154_round4.md](#) for why.

The strip beside each phone was a straight line, which meant registration had to be declared twice — once ending in a survey, once in a drawing — and the deck could only ever show one of them. A farmer looking at the log-in page could not see from the paper that creating an account, redeeming an invitation and resetting a password all start there.

FLOWS is a DAG now: a root and, for each screen, what it leads to. The deck lays it out left to right by depth, stacks siblings, and draws orthogonal elbows. That buys three things a line could not say — where the app **branches** (A3 into three), where it **converges** (both drawing routes finish on A11), and what is a dead end.

Five trees, the deepest eleven columns. Tiles are sized from that one so the deck keeps a single rhythm, and a tree that would run past the footer shrinks until it fits — with a build-time warning naming the flow if it still cannot.

Checks

`npm run check` · `npm run icons` · `npm run smoke` · `npm run deck`

478 renders across both roles, no console errors. The smoke test learned this round: the plot list must be lines rather than cards with the crop as a control; A9 offers no route until the type is answered and no route cards at all to a farm of trees; A12 offers no choices; and A10's drawing instruction must be one tap away in the guidance sheet rather than absent.

v1.5.4, fourth round of comment

Still **1.5.4**, on instruction. One deck, one number, while the review cycle is in flight; the specification underneath it has moved to **1.7**.

- Round 1, the recorded call: `Mockup_Changes_v154.md`
- Rounds 2 and 3, the first comments on the deck: `Mockup_Changes_v154_round2.md`
- Round 4, this document

The flow strip is a line again

Round 3 turned it into a tree. Round 4 turns it back, and the reason is worth writing down: a tree answers *what can this screen lead to*, which is a question about the app. A reviewer paging through a deck is asking *what was the farmer doing when he got here*, which is a question about one journey — and a diagram that shows every branch at once answers neither well on a printed page.

So `FLOWS` is a list of linear paths again, and each one now declares the

```
{ section: 'My Farm', name: 'Looking after the farm itself', ids: ['B2', 'B12'] }
{ section: 'My Plot', name: 'From the farm to one plot, and what is growing on it',
  ids: ['B2', 'B4', 'B5', 'B6'] }
{ section: 'Trees', name: 'From the farm to a tree group, and down to one tree',
  ids: ['B2', 'B13', 'B10'] }
```

B2 is on three of them. Which one prints beside it is decided by the section the page is filed under, so the copy of B2 in **My Farm** shows the farm journey and the copy in **My Plot** shows the plot journey. The reviewer's context picks the line, which is what "make sure the linear flow matches the current area in focus" asked for.

Fourteen paths across eight sections. Registration is two of them — one ending in a survey, one in a drawing — which is the price of a line, and a fair one.

Markers for the small buttons

"Add markers for small buttons that redirect to other slides, and small buttons that have important features behind them (like the map buttons). The markers should be located OUTSIDE of the phone mockup, and not cover any content."

A printout cannot be tapped. On paper an icon button that opens a whole screen and one that does nothing much look identical, and the three controls on B4's map — the reading picker, the date picker, the way into the map tab — are the clearest case: three glyphs, 40 dp each, carrying a third of what that screen can do.

in `app/ui/components.js`:

```
mapTool('scan', t('b4.openmap', 'Open in the map'), false, openMap, { deckTo: 'C1' })
helpButton(body, { title }) // declares its own note
overflowAction(onclick, label, { deckNote: 'Rename, edit the boundary, remove the plot' })
```


`deckTo` is the screen it leads to; `deckNote` is what it opens when it stays on this screen. Both render as `data-` attributes and **nothing appears in the app** — this is the same instruction as round 2's *no screen names another screen*, honoured rather than worked around. The attributes are invisible on a phone and read only by the deck builder.

`tools/screendeck.mjs` reads them off the rendered page with their positions and draws:

- a **numbered green disc in the gutter** — the 0.40 inch of white between the phone's right edge and the filmstrip, empty on every page of the deck — at the height of the control it names, so the eye travels straight across;
- a **numbered key** under the strip: `Open in the map → C1 Map (page 34)` for anything that leads somewhere, with the page number so the reviewer can turn to it, and the declared sentence for anything that opens in place.

Nothing is drawn over the phone and nothing is covered. Discs that would collide are pushed apart; the numbers still run top to bottom. Eight plot rows with the same chevron collapse to one marker rather than eight, and a marker pointing at a screen the deck does not print is dropped rather than sending the reviewer looking for a page that is not there.

DECK_OMIT

B11 was to come out of the slideshow. It was doing that by being left out of `SCREEN_GROUPS`, which quietly took it out of the app's own contact sheet too — a deck decision leaking into the app's index. `app/screens/index.js` now exports `DECK_OMIT = [ 'B11' ]`: the index stays complete, and the deck skips what the deck skips, in one named place.

Screen by screen

SCREEN	WHAT CHANGED
Global	Status vocabulary. Watch → Monitor , Action → Planned , with the meanings rewritten to say what to do: <i>No action required currently · Watch for changes and reassess · Complete as regular activity · Take action today</i> . The keys are unchanged, so nothing downstream had to move.
Global	Both calendars, everywhere. A date prints Gregorian and Hijri together — <code>12 February 2026 - 24 Sha'ban 1447</code> — ordered by the F8 setting (Gregorian first, Hijri first, or Hijri only). List cells and range endpoints stay short and Gregorian: two full dates either side of a dash is a paragraph where the column wants a stamp. B5's season bar was re-laid out for the extra width — the countdown moved above the bar and the two dates get a line each.
Global	Counts say urgent and count only urgent. The attention line counted everything that was not Good, which on any real farm says <i>something needs attention</i> every day of the year. It counts the urgent, and it is absent when there are none rather than printing "nothing urgent".
Global	A tree group is counted, not measured. Its hectares are the ground its parcels happen to cover — not what it is priced on, not what advice is calculated per, not a number anybody quotes about an orchard. Tree rows carry <code>1,240 trees</code> ; the hectares are gone from B13's header, from the B2 plot line and from A11.
A1	Register and Log in are the primary pair; the tour is a quiet filled container underneath rather than a third equal button.
A3	The code button reads "Send code to mobile number" .
A4 · A4A–A4D	The tour is five pages in the deck , one per panel, and sits after A14 — registration reads unbroken, then the detour that is offered on A1 as its own run.
FORGOT	Mobile and Email are the same switch as A3; the email route sends a reset link rather than a code, because that is what an email reset is.

SCREEN	WHAT CHANGED
A9	The fork — survey the whole farm, or draw my own plots — is offered only for field crops . A farm of trees is surveyed, because the tree count sets the price. Nothing is drawn disabled; pressing a route without a farm name puts the cursor in the name field and says why.
A10 / A10D	The drawing instruction lives behind a "How to draw this" chip. A10D gained the plot-type step it was missing. <i>(Round 5 moved the chip to an ⓘ on the app bar.)</i>
A11	Rectangles. The survey drew wobbling five- and six-sided parcels here after every other surface had been squared off. Same generator, same square grid, same look as B2 and the map. Tree rows show the number of trees, and the watch panel row is "Number of trees" .
A12	Retitled "Your survey" , cut to four short rows, warning box gone.
A13	Annual pricing on the plan cards at a 15% discount , the permission sentence on the trial card, and the bullets rewritten.
B2	The farm name at the top left is the picker, and it says so on hover; the plot list is hairlines rather than cards; only the chevron navigates.
B4	"See AI suggestions" → "See Advices" .
B11	Out of the deck — see <code>DECK_OMIT</code> above. Still in the app and still in the harness index.
B12	No longer re-offers the survey. <i>(Round 5 removed B12: it is A9B now.)</i>
F5 / F6	Carry the same annual discount as A13, from one exported constant.

Checks

`./tools/syntax.sh` · `npm run icons` · `npm run catalogue` · `npm run smoke` · `npm run deck`

494 renders across both roles, no console errors, 50 tiles for 50 screens on the contact sheet. The deck is 59 slides: cover, contents, 8 section dividers and 49 screens, 43 of which sit on one of the 14 paths.

v1.5.4, fifth round of comment

The last round on this version. Still **1.5.4**, on instruction; the requirement set underneath it is at **1.7**.

- Round 1, the recorded call: `Mockup_Changes_v154.md`
- Rounds 2 and 3: `Mockup_Changes_v154_round2.md`
- Round 4: `Mockup_Changes_v154_round4.md`
- Round 5, this document

Adding a farm is one flow, and it is an A flow

"B12 should really be an A screen, and we need to remove the name input. When the user adds a farm, either through the FIRST RUN or through the MY FARM screen, we should present A9, and then this new version of B12, that leads to A10 or A10D."

B12 was in the wrong place twice over. It was filed under **My Farm**, which made adding a farm look like something you do to a farm you already have; and it asked for the farm's name a second time, in a draft of its own, because it ran the fork without A9 in front of it. Two screens described the same two routes, and they had already drifted apart once.

So the flow is one flow now, from either end:

```
A9  name the farm · how you measure land · what is growing on it
    ↓ Continue
A9B survey the whole farm, or draw the plots yourself
    ↓
A10 (survey)   or   A10D (draw)
```

notices that qualify it — the ten-farm limit and the combined-plan offer, both of which appear only for an account that already has farms. B12 is gone from the registry, from the app map and from the deck.

Everything that adds a farm now opens A9: the registration walk, and **Add a farm** in the farm picker at the top left of My Farm. `startAddFarm()` is the one door. *Adding a plot* to a farm that already exists is a different thing and was being routed through the add-a-farm flow by accident; it has its own `startDrawPlot()` and goes straight to the canvas, because the farm is already named and its type already settled.

A9 has a Continue button

It had none, and the reason it had none is the reason it needed one: the two route cards *were* the action, so a farmer who had filled in a name, a unit and a crop type had nothing to press. With the fork moved to A9B, A9 ends the way every other form in the app ends.

Not disabled. A dimmed button does not say which answer is short — this one lands on whichever one is, and says why.

Screen by screen

SCREEN	WHAT CHANGED
A9	Asks only: the farm's name, the unit its land is measured in, and what is growing on it. Continue in the dock. The fork and the trees explanation have moved to A9B.
A9B	New — the old B12, minus its name field. Two route cards for a farm of field crops; for a farm with any trees on it, the reason there is only one way in and a single Draw my farm boundary button. Carries the ten-farm limit and the combined-plan note for an account that already has farms. <i>(Round 6 removed the trees branch: such a farm never reaches this screen.)</i>
B12	Removed.
A10 / A10D	The "How to draw this" chip is gone. The guidance is an ⓘ on the app bar, beside the subtitle that names what the screen is for — the same sheet, one tap, and the panel under the map is back to being about the plot rather than about the instructions. <code>helpChip()</code> has gone with it: there is one shape of help button in the kit now, not two.
A12	No card round We monitor / Priced by . Two facts in two lines did not need a box, and the box was the tallest thing on a screen whose whole point this round was that it had stopped being tall.

The deck

times over, which is a key repeated rather than a key.

produced a numbered disc the key could not explain, and a key line that began with a dash. The deck builder drops those now rather than drawing them, so a marker either says something or does not exist. Two related fixes went in with it: the ⓘ on a drawing screen names the screen it explains rather than saying

no-wrap setting introduced in round 4 is what made some of them come out blank in PowerPoint.

the cover, the contents and the eight section title pages leave the loop before any marker code runs. Worth saying plainly: on the generated deck I could not find a marker on the section 8 title page, in the file or in its XML. The guarantee is now structural rather than incidental, and if one is still showing in your copy, the slide number will tell me where to look.

Checks

```
./tools/syntax.sh · npm run smoke · npm run catalogue · npm run deck · npm run changelog
```

494 renders across both roles, no console errors, 50 tiles for 50 screens. The smoke test learned this round: A9 offers no route cards and must carry a Continue that refuses to proceed on an empty form; A9B is where the fork lives, gives a farm of trees no cards and a way on, and never asks for the farm name a second time; and A10's guidance is reachable from the **app bar** rather than from the body.

This document is generated: `tools/changelog.mjs` typesets these round records into `docs/Wafra_Mockup_Changes_v1.5.4.pdf`.

v1.5.4, sixth round of comment

Still **1.5.4**. The requirement set underneath it is at **1.7**.

- Round 1, the recorded call: `Mockup_Changes_v154.md`
 - Rounds 2 and 3: `Mockup_Changes_v154_round2.md`
 - Round 4: `Mockup_Changes_v154_round4.md`
 - Round 5: `Mockup_Changes_v154_round5.md`
 - Round 6, this document
-

A9B is for field crops, and it always offers both routes

"A9B should always show both options to detect farm (manual drawing vs farm boundary) and should only be shown if Crop is picked as type grown. Please make note of that in the slideshow, and in notes for this screen in the mockup — not on the screen itself."

Two rules read from the same end. Round 5 left A9B doing the branching itself: a farm of trees reached it and found one button and a paragraph explaining why the other option was missing, which is a screen apologising for existing. It does not exist for that farmer now.

carries the reason on its own last line — *"Trees are counted one by one from the imagery, so we read your whole farm from above. Next you will draw its boundary."* A9B is reached only by a farm of field crops, and because that is the only farm that reaches it, **both route cards are always there**. No state withholds one, greys one, or swaps one for an explanation.

A farmer never reads *"this screen appears when..."* — he either sees it or he does not. So it lives in the registry note in `app/screens/index.js`, on the A9B deck page as a green line between the title and the phone, and in that slide's speaker notes. The registry gained a `when` field for exactly this; any screen that is conditional can now say so to the deck without saying it to the farmer.

The deck

the registration walk for somebody who already has an account and the first screen of the way back in, and a reviewer opening **Log in** should not have to remember a page number from thirty pages earlier. The deck now prints a screen once per section it is filed in — same photograph, its own page, its own filmstrip: page 5 shows A3 on the signing-up path, page 22 shows it on

put two dark dividers between the farm and the plot you reach from it — and a plot is not a peer of the farm, it is what the farm contains. **My Farm** is now B2, then a plot and its cycles, then a tree group and one tree.

picking up B13 and B10. It was *Taking on another farm* for one round, which meant the busiest screen in the app illustrated the rarest thing anyone does on it. Adding a farm is A9 and A9B, it is drawn in First run, and it does not need drawing twice.

Screen by screen

SCREEN	WHAT CHANGED
A9	A farm with trees goes straight to A10 from here, and the last line says why. Field crops go to A9B.
A9B	Both routes, always. The trees branch and its single button have gone — that farmer never arrives. The condition is recorded in the registry note and printed on the deck page, not on the screen.
A13	Both Choose buttons are primary. Review S03 asked for the two to be the <i>same</i> , because a green button on one card and a grey one on the other is the app choosing for the farmer; sameness was the requirement and quietness never was, and a grey button under a price reads as the option you are being talked out of. Recorded as the one named exception to WF2.010's single primary action, so a <i>third</i> primary on A13 still fails the audit.
C1	Search is a tool, not a bar. The full-width pill across the top was covering the part of the map the farmer's own farm is in — it is the first glyph in the tool column now, where a search control belongs. The corner it vacated goes to the farm picker : which farm you are looking at, top left, over the map it names.
B2 · B13	The Open map button on the thumbnail was grey text on a blurred rectangle, which read as an unfinished caption rather than a control. It is a floating white pill with the glyph in a brand-tinted square and the arrow that says it leads somewhere — 32 dp of paint inside a 48 dp target, so it does not own the picture it sits on. One component, <code>openMapChip()</code> , shared by both screens.

A12 sounds like a person again

Cutting this screen down for length two rounds ago turned its prose into headlines — *"What you get for the boundary you just drew"*, *"And draw a boundary round each one"* — which is short, and cold, and this is the moment a farmer has just finished handing us his land and is waiting to hear what happens to it.

The lead is a sentence now: *"Thank you — we have your farm boundary. Here is what we will do with the land inside it, and what your price is based on."* And each of the three rows underneath is a whole sentence rather than a caption that only parses if you have read the title above it. Whole, but still short — warmth is not a licence to grow the screen back.

And the three steps are no longer in a card. They were `.row--static` inside a `card()`, which is the shape this app uses for the things you fill in or tap: a white panel, hairline dividers, 48 dp bands. Nothing on this part of the screen is either — it is us telling the farmer what we are about to do with his land while he waits for a price, and a settings-list frame around that reads as three switches he has failed to find. It is prose with a glyph beside it now: no box, no rules, no bands.

The deck note on **A9B** loses *"Both routes are always offered; neither is ever withheld."* The first sentence already says the screen is for field crops, and the code has no branch left that could withhold anything.

Checks

```
./tools/syntax.sh · npm run smoke · npm run catalogue · npm run deck · npm run changelog
```

494 renders across both roles, no console errors. The smoke test learned this round: a farm of trees is never shown the fork and A9 says why; a farm of crops reaches A9B and finds two live route cards, always; and the contact sheet is counted per **filing** rather than per screen, with a new check that no registered screen is filed in no section at all.

The deck is 58 slides: cover, contents, 6 section dividers and 50 screen pages across 49 screens.

v1.5.5, the comments on the v1.5.4 deck

The build moves to **1.5.5**. The requirement set underneath it is still at

- Round 1, the recorded call: `Mockup_Changes_v154.md`
- Rounds 2 and 3: `Mockup_Changes_v154_round2.md`
- Round 4: `Mockup_Changes_v154_round4.md`
- Round 5: `Mockup_Changes_v154_round5.md`
- Round 6: `Mockup_Changes_v154_round6.md`
- Round 7, this document

The comments arrived drawn onto the deck rather than as a document — peach note boxes, red frames, leader lines, strike-throughs and pasted screenshots, on twenty-two of the fifty-nine pages. **Forty-five changes**. Forty-four of them are in this build; the forty-fifth is at the end, and it is a question rather than a refusal.

A12 is deleted

"Delete? Not sure what purpose this screen is fulfilling. After A10 (Survey my whole farm), he should go to A11 (Survey results). It is too early for him to request a quote."

The whole screen was struck through, and so was its step in the flow ribbon. Both halves of what it did had already moved out from under it: the explanation of what a survey covers is now the tour's second panel, and the quote is asked for on A11, in front of the plots it is about.

So **A10 does the work it used to hand on**. Its button says *Request survey* — *"change to: 'Request survey'"* — and pressing it makes the farm record, runs the survey against the boundary just drawn, and puts up the pop-up the review asked for:

"Add a pop-up screen: We will notify you when the farm monitoring results are available (usually within one day)."

One sentence, one way onward, no Cancel — there is nothing to cancel, the work has been asked for. The button on it goes to **A11**, which is where the review said the farmer should be.

The pop-up is a new overlay shape. Every modal in the app was either a question with two answers or a picker; a plain acknowledgement had nowhere to live.

A11 carries the farm's own outline

Four notes landed on this page, and two of them are the same idea.

against the full-height map the farmer had been drawing on a screen earlier, so the plots the survey found were smaller than the single line he traced to find them. It is now 420 dp and the tallest thing on the screen. Not literally A10's height: A10 has nothing under its map, and a map that filled the phone here would hide the fact that there is a list of plots to approve.

"We should show the farm outline that the user selected in A10 as a reference point. If the user made a mistake, he should be able to adjust the farm outline to remove plots. An alternative way for him to remove plots."

A10 now stores the boundary on the farm, so the shape on this map is the shape he traced rather than a redrawing of it, and **every map in the app draws it** — which is the answer to a separate note on C1, *"on all the maps, let's show the farm boundary (if available)"*. A farm that arrived without a traced line — every fixture farm — gets the hull of its own plots with a little air round it, which is the second reading of *if available*: a farm with land has a shape even if nobody drew it.

Saving a corrected line takes every plot whose centre now falls outside it off the quote — the alternative way to remove plots the review named. Nothing is deleted; putting the line back puts them back.

crop.""* It follows *We found 8 plots inside your farm.*, and it is what makes Split on a row worth reaching for.

"Change to: 'What we will monitor'. Below that, we should show: Field crops 45.1 ha / Date palms 223 trees / Fruit trees 43 trees. If there is no value, we should show 0."

Three rows, always, with a nought where there is nothing — a row that vanishes when it is empty leaves the farmer to work out whether we found no palms or forgot to look. The app could not answer this before: the survey had one tree class and nothing under it. It now tells **date palms from other fruit trees**, which is a distinction the imagery can genuinely make, and it sits *under* the existing class rather than beside it — `kind` is still crops or trees, which is what the colours, the pricing and WF4.048's vocabulary are built on.

A13 loses three lines, and A13B appears

"Remove 'at all'." · "Move '+VAT' up one line to the right of 'month'." · "Delete."

The trial promise reads *"...and nothing is charged during the free trial"*: *at all* was doing the work of an argument in a sentence that is a promise, and a promise that protests is a promise being doubted.

caveat — instead of standing underneath looking like a second fact about the plan.

the farmer chooses a *level*; a second price for a billing period he has not been offered yet is a number to compare against the one he is deciding on.

Which is where the marker on page 13 comes in:

"Add screen Axx (slide 15)."

It sat in the gap between A13 and A14 — the one gap in the registration walk, and the screen three rounds of comment have called *the payment page* without ever drawing. **A13B · Confirm and pay** holds the two things that were waiting for it: the billing period, which is where a 15% saving is a reason rather than a distraction, and the permission A13 promises to ask for. Nothing is charged on it and it says so twice.

Both routes now finish the same way. The survey route used to drop the farmer straight into the app from A13, so one of the two ways in never saw **A14** at all. Choosing a level goes to A13B either way, and A13B to A14.

whether the app sounded as though it had taken possession of the farm.

The tour is six panels, and all of the words are the reviewer's

The five panels were placeholders held open for copy. The copy arrived, and it changed the shape of the thing: each panel now argues a part of the product, and a sixth was added second — "*new slide*" — because everything after it is a thing the service does and this is the sentence that says how it can.

1 of 6	<i>Enhancing your farm profitability through precision agriculture.</i> The body's last two clauses are switched: irrigation before fertiliser, which is the order the farmer meets them in.
2 of 6	<i>How our service works</i> — the new panel, over a satellite.
3 of 6	<i>Farm planner</i> , with D1 and F9 beside it.
4 of 6	<i>Irrigation and fertilization advice</i> , with D2 and D3.
5 of 6	<i>Optimizing crop yields</i> , with B5 and B6.
6 of 6	<i>Over 6 million farmers trust us worldwide</i> , three figures, and farm profitability underneath them.

and F9** could have been answered with six pasted PNGs and a note to re-paste them whenever a screen changed. They are the real screens instead, rendered small and cropped from the top the way an app-store shot is — so the tour cannot drift away from the app it is advertising. The thumbnails are inert and aria-hidden: a picture of a screen is not a control, and counting its buttons would fail the tour for touch targets that pass on their own pages.

The two panels with no screen to show draw their own artwork out of the same imagery the map screens use. Nothing here is a photograph, and nothing here needs maintaining separately from the app.

Advice, in the units the farmer works in

"Express water requirements in volumetric rates per plot (m³/ha) rather than simple millimeter depths."

D2 was already in cubic metres, and the figure it printed was already a rate — 8.4 mm/day of crop water use over a week at 85% efficiency is 693 m³ **per hectare**, and it was labelled as the plot's total. On a 137 ha block of palms that is a hundredfold error in the one number the farmer acts on. The headline is now the rate, said as a rate, with the plot's total and its area underneath: one figure to set the system by, one to check the bill against. Every weekly amount is in the same unit.

"Provide application rates in actual fertilizer product terms (e.g. kg/ha of Urea or NPK formulation) rather than elemental values alone."

D3 used to end its elemental block with a promise — *once you tell us which fertilisers you use, we'll show product equivalents too* — which asked the farmer for a shopping list before it would answer the question he came with.

amount for the plot, and more than one where more than one will do. The recommendation is still made in the nutrient, because that is what the crop is short of.

Compare plans is about the farm, not the satellite

"We don't show the practical features that are provided to the farmer under each plan. I don't think the satellite resolution, cloud-free data, etc. is useful. It suggests the basic service is degraded. ... Column 1: Features, Column 2: Basic plan, Column 3: Pro plan."

The satellite group is gone: it described our supply chain, and read as a list of the ways Basic is the cheap one. What is left is **four topics, each a thing the farmer does** — where his land is, what we watch on it, what we tell him to do, and what he can show afterwards — in **three columns**.

The previous shape printed every feature once, under *Basic includes* or *Pro adds*, which was the right fix for sixty rows of two identical ticks but meant a farmer reading the Basic column could not see that a feature was in both. A tick under each level says it outright, and where the levels differ by degree the cell carries the difference in words: *to 1 m* against *to 3 m*, *weekly amount* against *day-by-day schedule*.

The smaller marks

buttons was the one part of a deliberately tight screen doing nothing, and a farmer who wants to read about us before he registers now has somewhere to go. It lives in `brand.js` with the names, because re-labelling this app is replacing the artwork and those strings and nothing else.

opening block, shared as one component, because a farmer who cannot get past the front door is the person who needs it most. The buttons are quiet here and filled on F13: one primary action per screen, and on the front door that action is logging in.

word *us* from both buttons, and the footnote about contact details being loaded from our servers — a fact about our infrastructure told to somebody who wants help. The two buttons were 92 dp against the support ticket's 52; all three are now the same size.

"The example provided to the user should have a minimum of five corners. This will tell him that he is not limited to a perfect square. The shape should not be a perfect pentagon."

The v1.5.4 review had made it a rectangle on the argument that fields here are laid out in rectangles; this round reversed that on both screens, and ruled out the obvious way to satisfy it, because a regular polygon teaches its own wrong lesson — that corners come evenly spaced. What is there is a rectangle with one side stepped in and one corner cut, which is what a field bounded by a track and a neighbour looks like. The plot example also grew: *"the plot example seems small compared to the map area."*

delete 'For example: wheat, alfalfa...' — nine crops under an option nobody chooses *by* crop made the card four lines tall and invited the farmer to hunt for his own in a list that was never meant to be exhaustive. With them gone the map takes at least half the screen, which is the other note on that page:

quote*, the same words A11 now uses.

of the More section, as it makes the flow more natural, ending with F13.* It is where the profile sits on F0 itself — the card above every other row — so the printed walk now reads the way the screen does.

front of onboarding was answered on the slide it was written on: *"Romain, I'll explain to Hani that the order doesn't matter as they are parallel workstreams. No need to make changes."*

The one that is a question

"A9B should come before A9" — with the reason on the next page: "A9 only applies if user selects 'Survey my whole farm'."

A9 asks what is growing on the land, and that answer is what decides whether the fork exists at all. A farm with any trees on it never sees A9B: trees are counted one by one from the imagery, the count sets the price, and a farmer cannot trace that by hand — settled at the 22 August review, which asked for the fork to be shown to field crops alone. Asking the fork first would offer a date grower a route that ends in his being told he cannot take it, which is the screen apologising for existing that round 6 removed.

There is a reading of the note that works — ask *how shall we find your land* first, then ask the farm's details only on the survey route — but it needs the farm's name to move, and it needs a decision about what a tree grower sees. Worth five minutes rather than a silent reordering, so the pages are in the order the review asked for and the app is unchanged, with both facts written down beside the code in `app/screens/index.js`.

v1.5.6, the second pass over the same comments

The build moves to **1.5.6**. The requirement set underneath it is still at

- Round 7, the marks on the v1.5.4 deck: `Mockup_Changes_v155.md`
- Round 8, this document

Eleven notes on what round 7 produced. Two of them take work back out, and the rest are about the size and the substance of pictures.

The map is one size, on all three screens

"For A10, A10D and A11 — let's make the map the same size, 65% of the screen, no margins left, right or top. Scroll up / down to show the rest."

The farmer draws his farm boundary on A10, draws his plots on A10D and reads what came back on A11, and until now those were three different maps of three different sizes: A10's filled everything between the bar and the button, A10D's took what the panel under it left, and A11's was a card with a radius and a margin. A map that changes shape between the drawing and the reading has to be re-read each time.

All three now use `mapBand()` — 65% of the phone, flush to the top and to both edges, with whatever is under it scrolling up over nothing. The percentage is the reason the band is a **direct child of the scroll area** rather than a block inside the page: a percentage height resolves against the nearest ancestor that has a definite one, and `.page` does not. So those three screens return an array as their body, the band first.

one warning that can appear: 65% leaves a band of empty paper above the button. That is the note followed literally, and it is worth a word at the next review — the alternative is to let A10's map take the room nothing else wants, at the price of the three maps no longer matching.

A11 is Survey results, and the boundary control is on the bar

"Move the 'adjust farm boundary' to the top bar, right side, as a smaller, subtle button." · "Change slide title and screen name to Survey Results."

The boundary control was a full-width row under the map, which gave a correction the same weight as the thing it corrects. It is now an icon and a word on the right of the app bar, where every other per-screen action in the app already lives. `barAction()` gained a `title`, so the bar reads **Boundary** and a screen reader still hears *Adjust the farm boundary* — the same control, and not the same length.

The screen is **Survey results** on the page and in the deck. *Summary of plots to be monitored* described the list rather than naming the screen, which left the reviewer's own shorthand — he calls it the survey results throughout — nothing on screen to attach to. It is set in sentence case to match every other screen name in the deck.

A13B is withdrawn

"Remove A13b — it wasn't there before and shouldn't be there now."

The marker on page 13 of the v1.5.4 deck — *"add screen Axx (slide 15)"* — was read as a request for the payment page. It was not. The screen is gone, and

stated once, in *Before you buy* at the foot of A13, where the other commercial facts are.

Both routes still finish on A14, which is the part of round 7 that stands: the survey route used to drop the farmer into the app from A13 and never see the screen that says the work is done.

The tour shows pictures

"Screen A4A — use the satellite picture attached instead of your own drawing." · "A4E — use the image of the crops attached." · "A4B to A4D — use actual pictures of the screen, generated, cropped, with light border."

Both photographs were already in the repository's own history: the reviewer had pasted them onto slides 17 and 21 of the deck he marked up, so they came out of that file at full size rather than being redrawn. They are in `app/imgs/tour/` as AVIF, 42 KB and 134 KB.

The middle three panels used to mount the real screens live and scale them down. It kept the tour honest by construction and cost three things this answer removes: a page carrying three primary buttons and sixty sub-36 dp touch targets belonging to other screens, six extra renders every time the tour opened, and — on paper — a photograph of a photograph.

The honesty is a build step now. `npm run tourshots` drives the app, photographs D1, F9, D2, D3, B5 and B6 at twice the phone's pixels, crops each to the top 62% and writes it into `app/imgs/tour/`. It is the first thing to run when one of those six screens changes. The pictures are checked in, because this app has no build step and `index.html` has to open from a `file://` URL.

Each sits behind a hairline border and a soft shadow — the review's *light border* — because they are pictures of a white app on a white panel and without an edge they float.

The smaller marks

contact buttons to the bottom of the screen."* They sat directly under the two links, which put a help offer in the middle of a short screen where it read as a third way to log in. A `page--fill` page and a spacer put it against the bottom on a tall phone and let it simply follow the form on a short one — pushed down, never pushed off.

which sized them to a share of the dock and then let the labels wrap:

`actionDockPair` is for, and a pair does not wrap.

A9B goes back after A9, and says why on the page

"A9B: let's keep it AFTER A9 in the slides (against comments), and add a visible note to the powerpoint explaining it."

Round 7 printed the deck in the order the first pass asked for and left the app alone, which put the pages and the flow ribbon on the same page disagreeing with each other. The pages are back in the order the app walks, and **A9B's page now carries the reason in the green band under its title** — the band that already carries the condition under which the screen appears at all:

Shown only when the farm's type is FIELD CROPS. A farm with trees goes from A9 straight to A10. PRINTED AFTER A9, AND NOT BEFORE IT: the 01/09 note asks for A9B first, but A9 is where the farmer says what is growing, and that answer is what decides whether this screen appears at all — so the fork cannot be asked before it. Open for discussion; see the change record.

Which is better than the reordering was. A page out of order says nothing about why; a note on the page does.